

CLOMETRO 7HP

Name of Drug

CLOMETRO 7HP is a combination pack containing Claritrox (active ingredient clarithromycin) 250mg tablet, Metgyl (active ingredient metronidazole) 400mg tablet and Omilock (active ingredient omeprazole) 20 mg capsule.

Description

Clarithrox 250 mg tablet – A yellow colour, modified capsule shaped, SM embossed with break-line on one side, film coated tablet.

Metgyl 400 mg tablet - Round, yellow colour, film coated tablets.

Omilock 20 mg capsule - Caramel/Pink coloured hard gelatin capsule, size “2” containing off-white granules.

Pharmacology

Helicobacter pylori (H. pylori) is a gram-negative spiral, flagellated, micro-bacterium. It is found to colonise and congregate at the antrum of the stomach, its natural habitat being the gastric mucosa. H. pylori has been found to be the major cause of duodenal and gastric ulcer diseases which are not cured by non-steroidal / anti-inflammatory drug (NSAID) agent. Eradication of H. pylori is therefore an appropriate therapy for peptic ulcers and has been successfully achieved in approximately 97% of patients, after a therapy of 7-day regime with clarithromycin, metronidazole and omeprazole.

Clarithrox 250 mg tablet

Clarithromycin is a semisynthetic macrolide antibiotic. Clarithromycin binds to the 50S ribosomal subunit of the 70S ribosome of susceptible organisms, thereby inhibiting bacterial RNA-dependent protein synthesis. Clarithromycin is active in vitro against a variety of aerobic and anaerobic gram positive and gram-negative microorganisms. Additionally, the 14-OH clarithromycin metabolite also has clinically significant antimicrobial activity.

Metgyl 400 mg Tablet

Antibacterial (systemic); antiprotozoal – Microbicidal; active against most obligate anaerobic bacterial and protozoa by undergoing intracellular chemical reduction via mechanisms unique to anaerobic metabolism. Reduced metronidazole, which is cytotoxic but short-lived, interacts with bacterial DNA to cause a loss of helical structure, strand breakage, and resultant inhibition of nucleic acid synthesis and cell death.

Omilock 20 mg Capsule

Omeprazole is activated at an acidic pH to a sulphenamide derivative that binds irreversibly to H⁺, K⁺-ATPase, an enzyme system found at the secretory surface of parietal cells. It thereby inhibits the final transport of hydrogen ions (via exchange with potassium ions) into the gastric lumen. Since the H⁺, K⁺-ATPase enzyme system is regarded as the acid (proton) pump of the gastric mucosa, omeprazole is known as a gastric acid pump inhibitor. Omeprazole inhibits both basal and stimulated acid secretion irrespective of the stimulus.

It is conclusively evident from the mechanism action of the above three drugs simultaneously administered offers the most effective treatment of H pylori. Therefore the combination pack provides the three drugs readily to the patient.

Pharmacokinetics

Clarithrox 250 mg Tablet

Absorption: Well absorbed from the gastrointestinal tract; stable in gastric acid; food delays the rate, but not the extent, of absorption; bioavailability is approximately 55% in healthy volunteers.

Distribution and Elimination: Widely distributed into tissue and fluids, high concentrations found in nasal mucosa, tonsils, and lungs; concentrations in tissue are higher than those in serum because of high intracellular concentrations.

Binding to Plasma Proteins: 65 to 75%.

Biotransformation: Hepatically metabolised.

Half - life: Normal renal function - 250 mg every 12 hours – 3 to 4 hours.

Renal function impairment (creatinine clearance of < 30 ml per minute) – Approximately 22 hours.

Peak serum concentration: Clarithromycin: Steady-state:

250 mg every 12 hours: approximately 1 mcg/mL.

14 - Hydroxylclarithromycin: Steady- state:

250 mg every 12 hours: approximately 0.6 mcg/mL.

Elimination: Renal – Approximately 20 and 30 %, respectively, of the dose of 250-mg tablets given twice a day is excreted in the urine as unchanged drug. Faecal – Approximately 4 % of a 250-mg dose is excreted in the faeces

Metgyl 400 mg Tablet

Absorption - Well absorbed orally; bioavailability at least 80%

Distribution - Distributed to saliva, bile, seminal fluid, breast milk, bone, liver and liver abscesses, lungs, and vaginal secretions; crosses the placenta and blood brain barrier, also.

Volume: In adults: Approximately 0.55 L/kg.

Protein binding - Low (<20 %)

Biotransformation – Hepatic

Half-life - In adults - Normal liver function: 8 hours

Time to peak concentration - 1 to 2 hours (oral)

Elimination - Renal – 60 to 80%; of this amount, approximately 20% excreted unchanged in urine.

Omilock 20 mg Capsule

Absorption - Rapid; 50 – 65%.

Distribution - Distributed in tissue, particularly gastric parietal cells.

Protein binding - Very high, approximately 95% bound to albumin and alpha₁-acid glycoprotein.

Biotransformation - Hepatic, extensive.

Half-life

Plasma – Normal hepatic function – 30 minutes to 1 hour.

Onset of action - Within one hour.

Time to peak concentration - Within 30 minutes to 3.5 hours.

Time to peak effect - Within 2 hours.

Duration of action - Up to 72 hours or more (96 hours required for full restoration of acid production).

Elimination

Renal – 72 to 80%.

Fecal – 18 to 23%.

In dialysis – Not readily dialyzable, because of extensive protein binding.

Indication

Treatment of peptic ulcers associated with *Helicobacter pylori* infection.

Contraindications:

Claritrox 250 mg Tablet

Clarithromycin is contraindicated in patients with a known hypersensitivity to clarithromycin, erythromycin, or any of the macrolide antibiotics. Concomitant administration of clarithromycin with cisapride, pimozide, or terfenadine is contraindicated. Clarithromycin and ergot derivatives should not be co-administered.

Metgyl 400 mg Tablet

During the first trimester of pregnancy and during lactation; Patients known to be sensitive or hypersensitive to nitroimidazoles; Organic neurological disorders (CNS disorders); Blood dyscrasias or other anomalies of the blood picture; Concurrent use ethyl alcohol; Severe hepatic failure.

Omilock 20 mg Capsule

There are no known contraindications to the use of Omilock Capsule.

Risk-benefit should be considered when the following medical problem exist:

Sensitivity to Omeprazole.

Side Effects / Adverse Reactions:

Claritrox 250 mg Tablet

The most frequently reported events in adults were diarrhoea, nausea, abnormal taste, dyspepsia, abdominal pain/discomfort, and headache. Most of these events were described as mild or moderate in severity. Of the reported events, only 1 % was described as severe. Allergic reactions ranging from urticaria and mild skin eruptions to rare cases of anaphylaxis and Steven Johnson's syndrome have occurred. Other spontaneously reported events include glossitis, stomatitis, oral moniliasis, vomiting, tongue discoloration, and dizziness. Transient CNS events including anxiety, behavioural changes, confusional states, depersonalisation, disorientation, hallucinations, insomnia, nightmares, psychosis, tinnitus and vertigo have been reported. Hepatic dysfunction, including increased liver enzymes and hepatocellular and /or cholestatic hepatitis, with or without jaundice, has been infrequently reported. Rarely, clarithromycin have been associated with ventricular arrhythmias, including ventricular tachycardia and torsades de pointes, in individuals with prolonged QT intervals. Pseudomembranous colitis has been reported rarely with clarithromycin, and may range in severity from mild to life threatening. There have been report of pancreatitis and convulsion.

Metgyl Tablet 400 mg

The adverse effects of metronidazole are generally dose-related. The most common are gastro-intestinal disturbances, especially nausea and an unpleasant metallic taste; nausea sometimes accompanied by headache, anorexia, and vomiting. Diarrhoea, dry mouth, a furred tongue, glossitis, and stomatitis may also occur. Peripheral neuropathy, usually presenting as numbness or tingling in the extremities, and epileptiform seizures are serious adverse effects on the nervous system that have been associated especially with high doses of metronidazole or prolonged treatment. Weakness, dizziness, ataxia, drowsiness, insomnia and changes in mood or mental state such as depression or confusion have also been reported. A moderate leucopenia has been reported in some patients but the white cell count has always returned to normal before or after treatment has been completed. Skin rashes and pruritus occur occasionally and anaphylaxis has been reported rarely. Other side effects include urethral discomfort and darkening of the urine. Raised liver enzyme values have occasionally been reported.

Omilock 20 mg Capsule

Omeprazole capsule is well tolerated and adverse reactions have generally been mild and reversible. The following side / adverse effects have been selected on the basis of their potential clinical significance (possible signs and symptoms in parentheses where appropriate) – not necessarily inclusive:

Those indicating need for medical attention

Incidence rare

Hematologic abnormalities, specifically anemia (unusual tiredness or weakness) ; eosinopenia; leukocytosis (sore throat and fever); neutropenia (continuing ulcers or sores in mouth); pancytopenia or thrombocytopenia (unusual bleeding or bruising); hematuria (bloody urine); proteinuria (cloudy urine); urinary tract infection (bloody or cloudy urine; difficult, burning, or painful urination; frequent urge to urinate)

Those indicating need for medical attention only if they continue or are bothersome

Incidence more frequent

Abdominal pain or colic

Incidence less frequent

Asthenia (muscle pain; unusual tiredness); central nervous system (CNS) disturbances, specifically dizziness, headache, somnolence (unusual drowsiness); or unusual tiredness; chest pain; gastrointestinal disturbances, specifically acid regurgitation (heartburn); constipation; diarrhea or loose stools; flatulence (gas); or nausea and vomiting; skin rash or itching, urticaria and pruritus, paresthesia, insomnia and vertigo. Blurred vision, taste disturbance, peripheral oedema, increased sweating, gynaecomastia, bronchospasm, encephalopathy in patients with preexisting severe liver disease, hepatitis with or without jaundice, rarely interstitial nephritis and hepatic failure.

Increase in liver enzymes have been observed.

Precautions / Warnings:**Claritrox 250 mg Tablet**

Cross sensitivity and/or related problems

Patients who are hypersensitive to erythromycin or other macrolides may also be hypersensitive to clarithromycin.

Hepatic and renal function impairment

Clarithromycin is principally excreted by the liver. Therefore, caution should be exercised in administering the antibiotic to patients with impaired hepatic function. Caution should also be exercised when administering clarithromycin to patients with moderate to severe renal failure.

Pseudomembranous colitis has been reported with nearly all anti-bacterial agents, including clarithromycin, and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhea subsequent to the administration of anti-bacterial agents. Prolonged or repeated use of clarithromycin may result in an overgrowth of non-susceptible bacterial or fungi. If super infection occurs, clarithromycin should be discontinued and appropriate therapy instituted.

Metgyl 400 mg Tablet

Metronidazole should be used with great care in patients with blood dyscrasias or with active disease of the central nervous system. All patients receiving metronidazole for more than 10 days should be monitored and treatment discontinued if signs of peripheral neuropathy or CNS toxicity develop. Doses should be reduced in patients with severe liver disease.

Omilock 20 mg Capsule

When gastric ulcer is suspected, the possibility of malignancy should be excluded before treatment with Omilock is instituted, as treatment may alleviate symptoms and delay diagnosis.

Risk-benefit should be considered when the following medical problems exist:

Hepatic disease, chronic, current or history of (dosage reduction may be required due to increased half-life in chronic hepatic disease).

Pregnancy and Lactation**Claritrox 250 mg Tablet**

Pregnancy: There are no adequate and well-controlled studies in pregnant women. Clarithromycin should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Breast Feeding - Clarithromycin and its active metabolite are distributed into human breast milk. Therefore, caution should be exercised when clarithromycin is administered to a nursing woman.

Metgyl 400 mg Tablet

Pregnancy / Reproduction - Studies in rats have not shown that metronidazole cause impaired fertility or birth defect in the fetus. When administered orally, no foeto-toxicity was seen in pregnant mice. However, the use of metronidazole in the treatment of trichomoniasis is not recommended during the first trimester.

Breast-feeding - Use is not recommended in nursing mothers. During treatment with metronidazole, the breast milk should be expressed and discarded. Breast-feeding may be resumed 24 to 48 hours after treatment is completed.

Omilock 20 mg Capsule

Pregnancy - Adequate and well-controlled studies in humans have not been done and the drug is not recommended.

Breast-feeding - It is not known whether omeprazole is distributed into human breast milk. However, because omeprazole has been shown to cause tumorigenic and carcinogenic effects in animals, a decision should be made on whether nursing should be discontinued or the medication withdrawn, taking into account the importance of the omeprazole to the mother.

Drug Interactions:**Claritrox 250 mg Tablet**

Carbamazepine - (administration of carbamazepine with clarithromycin has been shown to increase significantly the plasma concentration of carbamazepine; carbamazepine serum levels should be monitored)

Digoxin - (concurrent administration of digoxin with clarithromycin has been shown to increase serum digoxin concentrations; monitoring of digoxin serum levels is recommended)

Rifabutin or Rifampin - (concurrent use of clarithromycin and rifabutin or rifampin decreases the serum concentration of clarithromycin by > 50 %)

Terfenadine - (concurrent use of clarithromycin and terfenadine may increase the plasma concentration of terfenadine and its active metabolite by 2 to 3 times; concurrent use should be avoided)

Theophylline - (concurrent administration with clarithromycin has been shown to increase the AUC of theophylline by 17 %; monitoring of theophylline serum levels is recommended)

Warfarin - (concurrent administration with clarithromycin has been shown to potentiate the effects of warfarin; prothrombin time should be closely monitored)

Ergotamine or dihydroergotamine - (concurrent use with clarithromycin has been associated in some patients with acute ergot toxicity)

Triazolam - (There have been reports of CNS effects with the concomitant use of clarithromycin and triazolam)

Cyclosporine, tacrolimus, hexobarbital, phenytoin, alfentanil, Quinidine/disopyramide, lovastatin, bromocriptine, valproate, astemizole and other drugs metabolised by the cytochrome P450 system - (concurrent use with clarithromycin may be associated with elevations in serum levels of these other drugs; serum concentrations should be closely monitored)

Drug metabolised by CYP3A isoenzyme - Clarithromycin has the potential to interact with the large number of drugs through its action on hepatic cytochrome P450 isoenzymes. Such interactions can result in severe adverse effects, including ventricular arrhythmias with the non-sedative antihistamines astemizole and terfenadine and with the prokinetic drug cisapride.

Zidovudine - Simultaneous oral administration of zidovudine and clarithromycin resulted in a lower peak serum concentration [C_{max}], lower AUC, and delayed time to peak concentration [T_{max}] of zidovudine should be taken at least 4 hours apart.

Metgyl 400 mg Tablet

Alcohol - It is recommended that the metronidazole not be used concurrently with ingestion of alcohol may resulting in disulfiram-like effects.

Anticoagulants, coumarin- or indandione-derivative - Effects may be potentiated when these agents are used concurrently with metronidazole, periodic prothrombin time determinations may be required to determine if dosage adjustments anticoagulants are necessary.

Cimetidine - Hepatic metabolism of metronidazole may be decreased when metronidazole and cimetidine are used concurrently, possibly resulting in delayed elimination and increased serum metronidazole concentrations.

Disulfiram - It is recommended that metronidazole not be used concurrently with, or for 2 weeks following, disulfiram in alcoholic patients; such use may result in confusion and psychotic reactions.

Lithium - Lithium concentrations may increase when metronidazole therapy is introduced; serum lithium and serum creatinine levels should be monitored several days after beginning metronidazole in order to detect impending lithium intoxication.

Phenobarbital - Phenobarbital may induce microsomal liver enzymes, increasing metronidazole's metabolism and resulting in a decrease in half-life and plasma concentration.

Phenytoin - Metronidazole may impair the clearance of phenytoin, increasing phenytoin's plasma concentration.

Aspartate amino transferase assay - It may give spuriously low values in patients taking metronidazole depending on the method used.

Omilock 20 mg Capsule

Digoxin - Simultaneous treatment with omeprazole and digoxin in healthy subjects lead to a 10% increase in the bioavailability of digoxin as a consequence of the increased intragastric pH.

Diazepam, Phenytoin and Warfarin - Omeprazole inhibits the metabolism of drugs metabolised by the hepatic cytochrome P450 enzyme system and may increase plasma concentrations of diazepam, phenytoin, and warfarin. Interaction with other drugs also metabolised via cytochrome P-450 system cannot be excluded.

Recommended Dosage, Dosage Schedule and Route of Administration:

Clarithromycin Tablet 250 mg twice daily, Omilock Capsule 20 mg twice daily, and Metgyl Tablet 400 mg twice daily, all to be taken before meals for 7 days, and can be prolonged to up to two weeks maximum as per the recommendation of the doctor or physician.

Symptoms and Treatment for overdose and antidotes:

Claritrox 250 mg Tablet

Reports indicate that the ingestion of large amounts of clarithromycin can be expected to produce gastrointestinal symptoms. One patient who had a history of bipolar disorder ingested 8 g of clarithromycin and showed altered mental status, paranoid behaviour, hypokalemia and hypoxemia. Allergic reactions accompanying overdosage should be treated by the prompt elimination of unabsorbed drug and supportive measures.

Metgyl 400 mg Tablet

Symptoms

One case report of a voluntary overdose of 4200 mg in a 16-year-old pregnant woman reported that the patient developed disorientation, which resolved without specific treatment. Larger doses than this reported overdose have been given to patients as a radiosensitizer without severe toxicity.

Treatment:

Since there is no specific antidote, treatment for metronidazole overdose should be symptomatic and supportive.

Omilock 20 mg Capsule

None has been reported; Single oral doses of up to 160 mg and total oral doses of up to 300 mg/day have been given without adverse effects.

Symptoms / Clinical Effects of overdose

The following effects have been selected on the basis of their potential clinical significance: Blurred vision; confusion; diaphoresis (increased sweating); drowsiness; dryness of mouth; flushing; headache; nausea; tachycardia (fast or irregular heartbeat)

Treatment

Since there is no specific antidote for overdose with omeprazole, treatment should be symptomatic and supportive. Due to extensive protein binding, omeprazole is not readily dialyzable. Patients in whom intentional overdose is confirmed or suspected should be referred for psychiatric consultation.

Packing / Pack Sizes:

CLOMETRO 7HP combination pack consists of 14 Claritrox 250 mg tablets (blister pack), 14 Metgyl 400 mg tablets (blister pack) and 14 Omilock 20 mg capsule (blister pack)

Storage Conditions, User Instructions and Pharmaceutical Precautions:

Store below 30°C. Protect from light.

Shelf life: 3 years

Name and Address of Manufacturer and Distributor in Malaysia:

SM PHARMACEUTICALS SDN BHD (218620-M)

Lot 88, Sungai Petani Industrial Estate

08000 Sungai Petani

Kedah Darul Aman, Malaysia.

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